Experiment:- 9

Problem statement:-

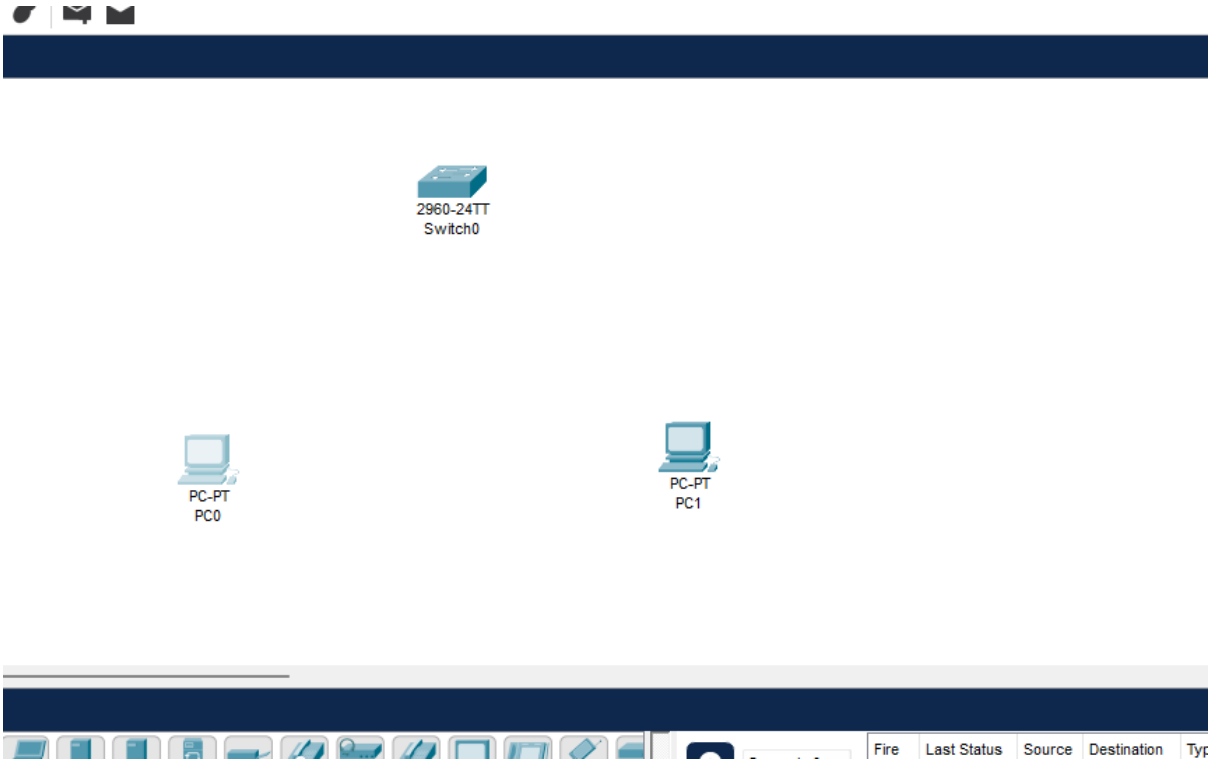
Implement a transport layer simulation to demonstrate process-to-process communication using UDP, TCP, and SCTP protocols. Compare these protocols in terms of connection establishment, data transmission, and congestion control using Cisco Packet Tracer.

Aim:- **Transport Layer Simulation: UDP, TCP, and SCTP Comparison**

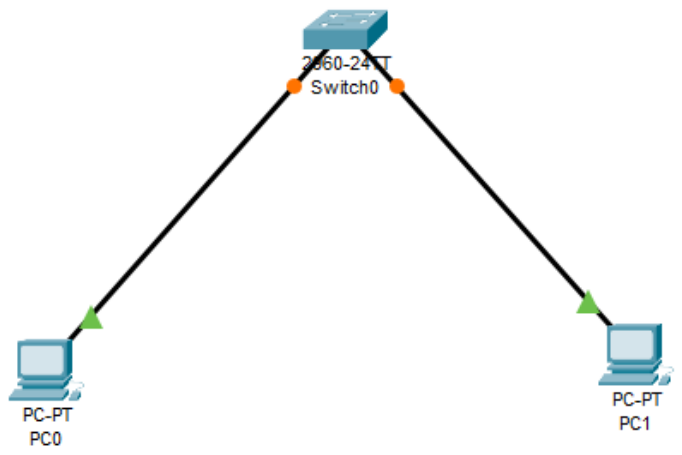
Procedure:-

Step 1: Setup the Network Environment

1. Open Cisco Packet Tracer.
2. Create a new project.
3. Add the following devices:
   - Two PCs (PC0 and PC1)
   - One Switch (2960)

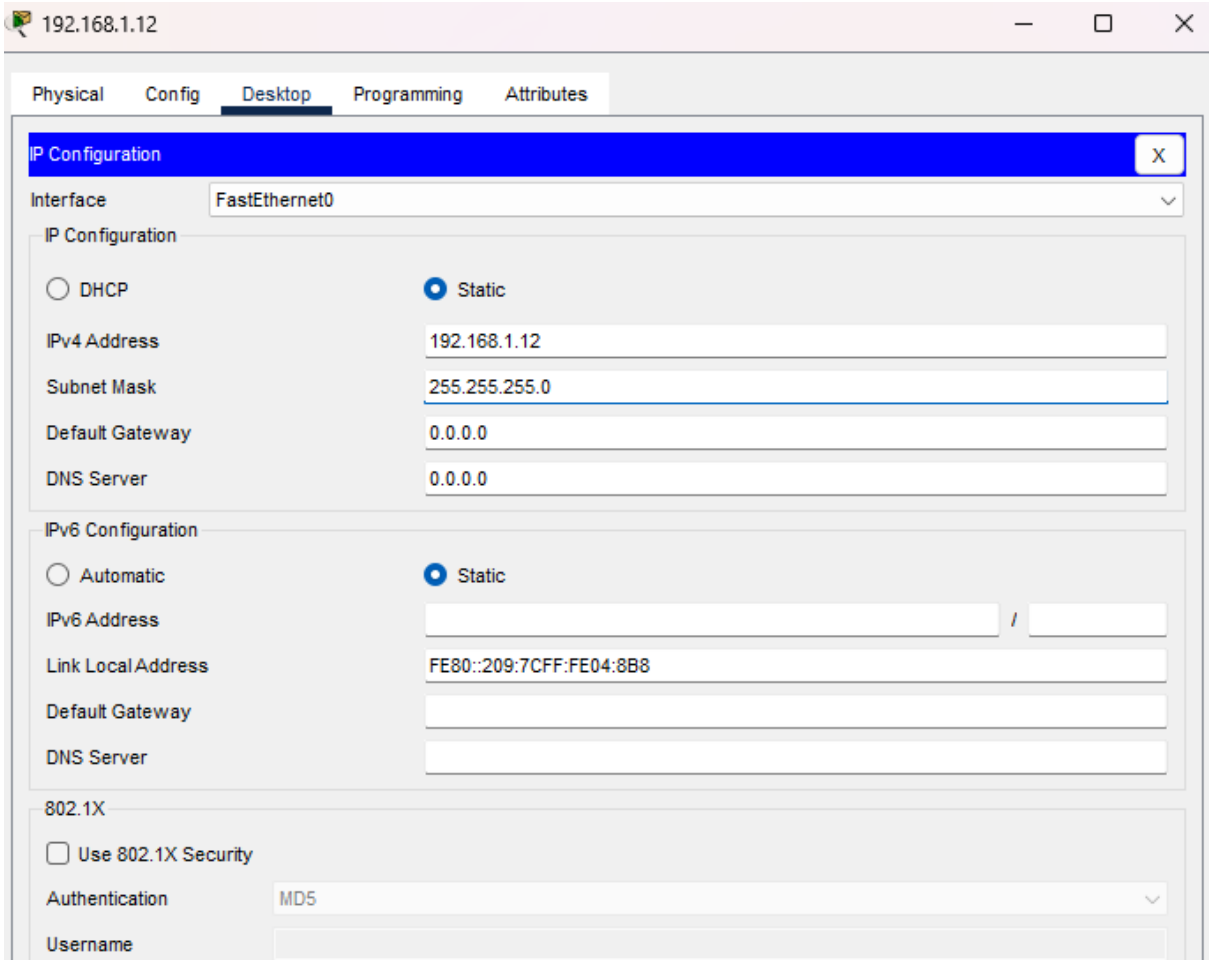


4. Connect:
   - PC0 → Switch
   - PC1 → Switch
   using Copper Straight-Through cables.



◈ Step 2: Configure IP Addresses

- **PC0 PC1:**

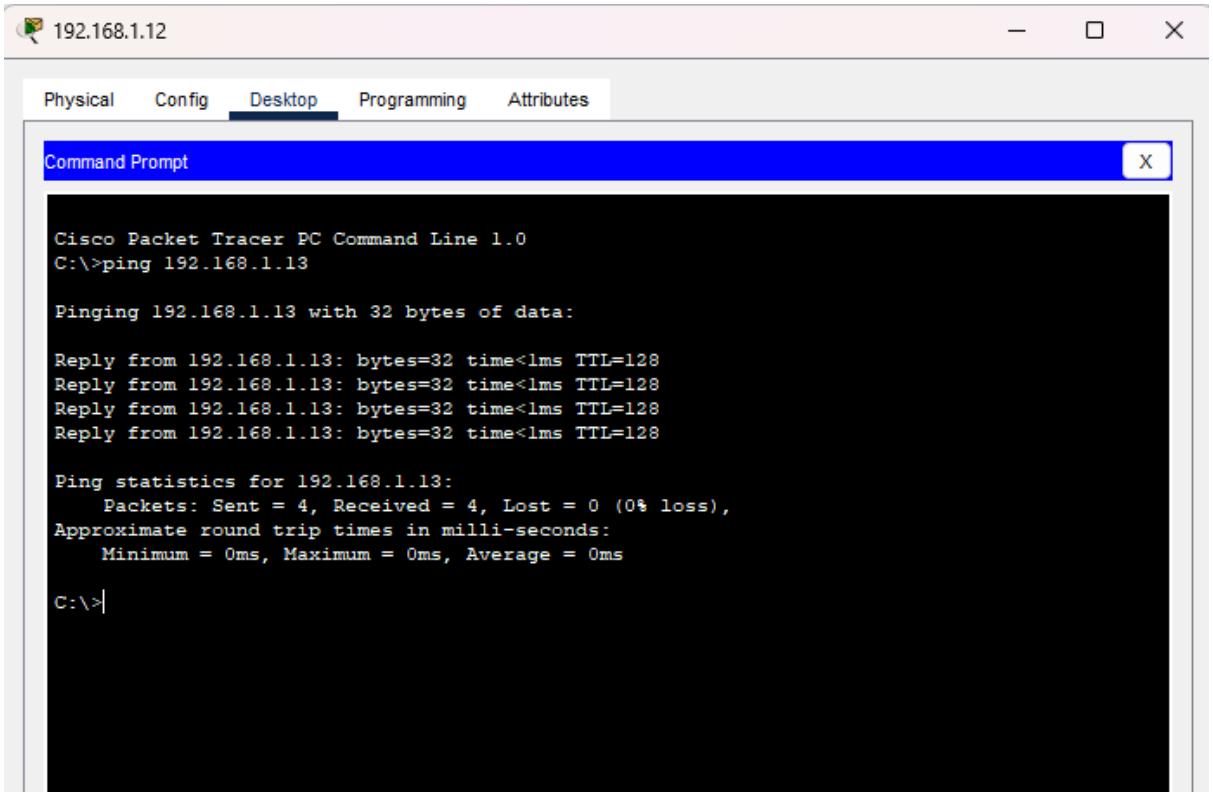


◈ Step 3: Test Connectivity

1. Open Command Prompt on PC0.
2. Type:

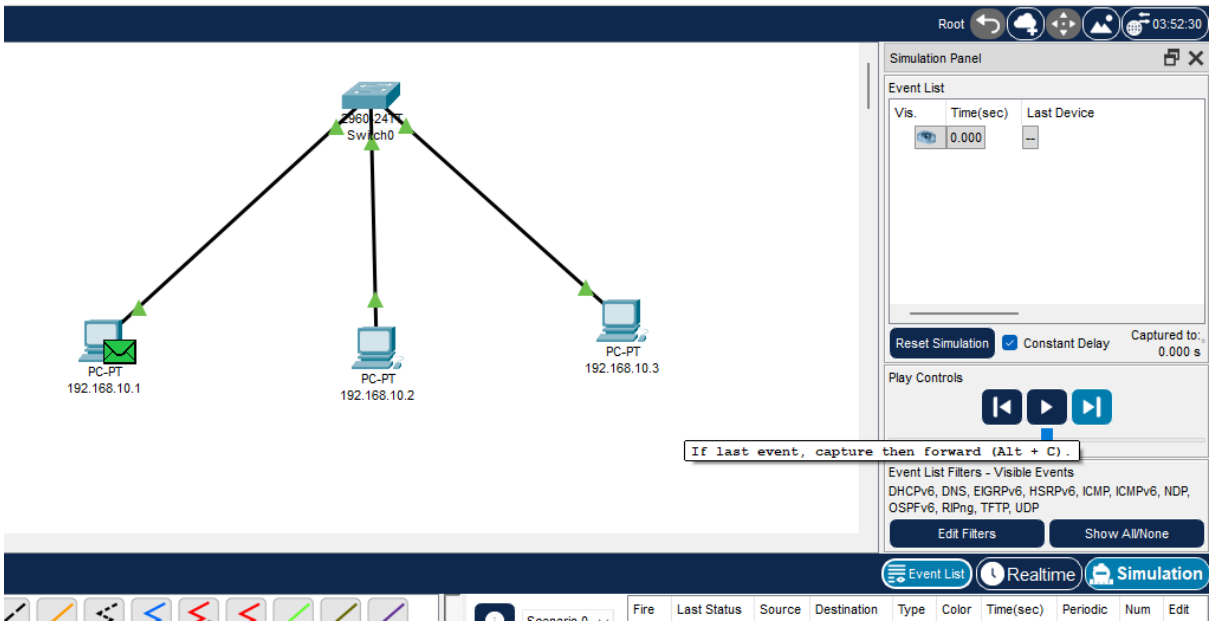
```
ping 192.168.1.2
```

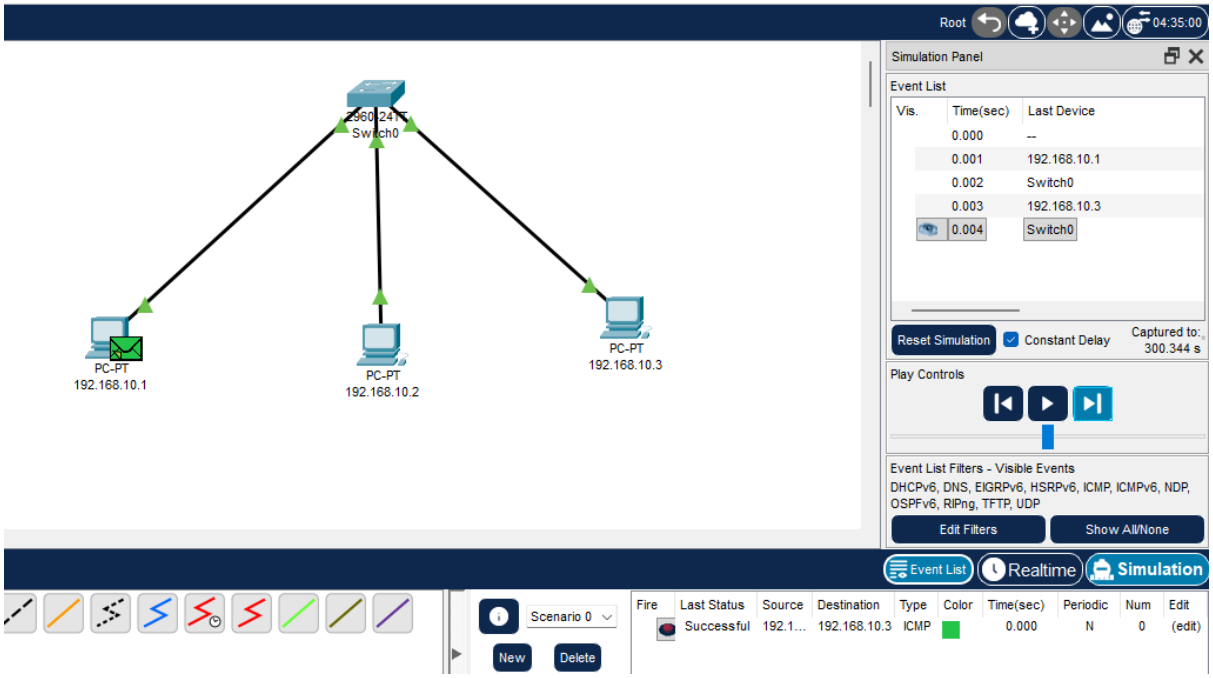
3. Successful replies indicate proper network setup.



UDP Communication:-

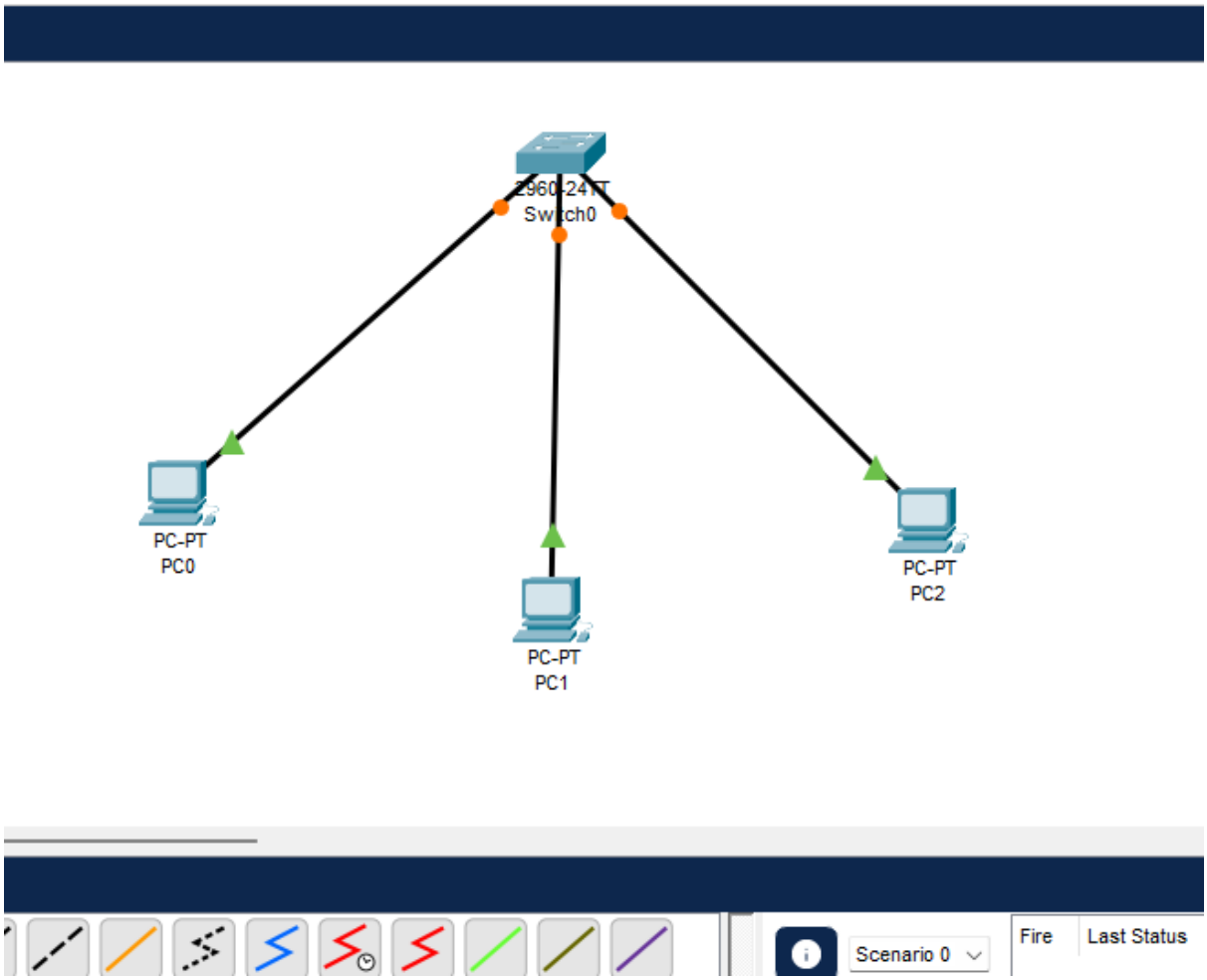
1. Switch to **Simulation Mode**.
2. Select **Add Simple PDU (envelope icon)**.
3. Click on:
    - PC0 (source)
    - PC1 (destinatio0
    -



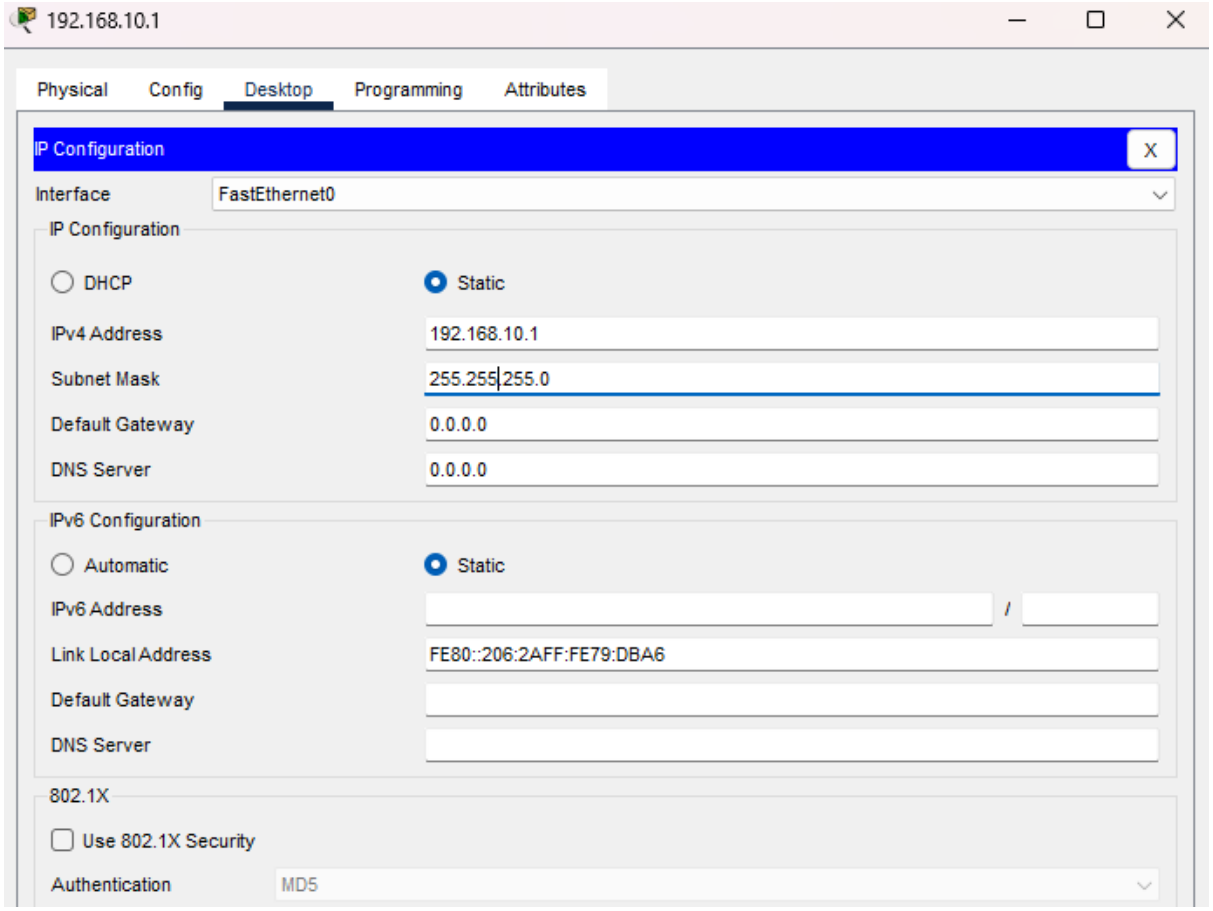


Simulate TCP Communication:-

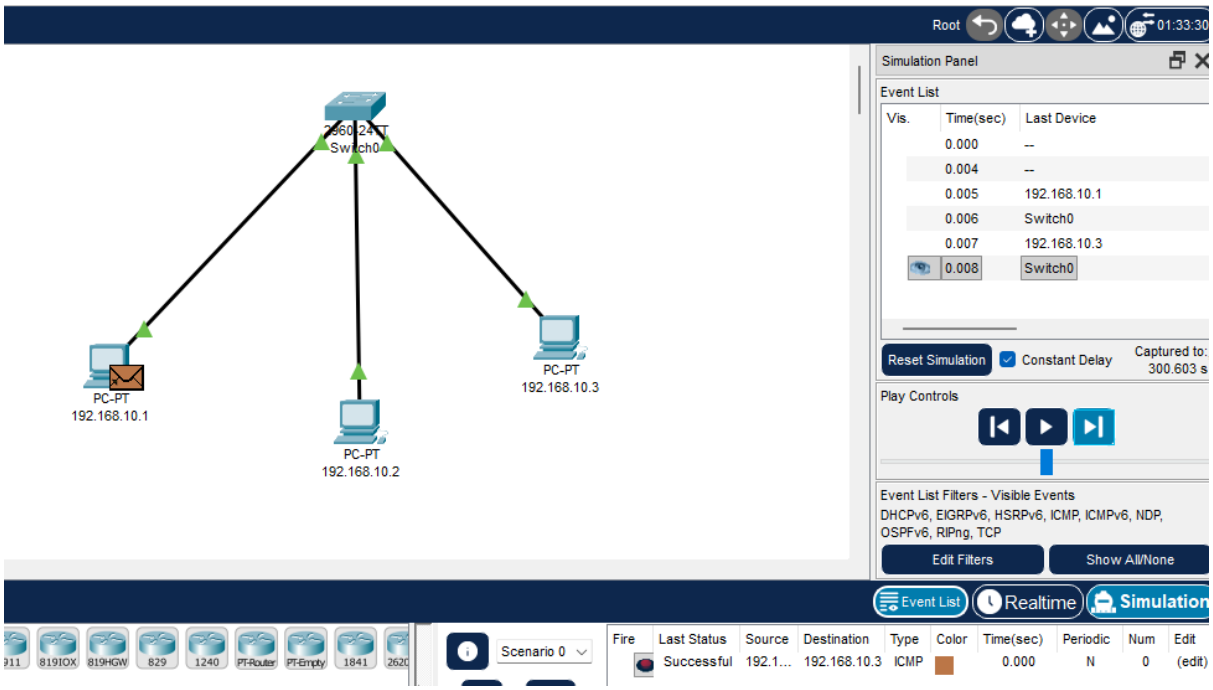
1. Open **Cisco Packet Tracer** and create a new project.
2. Add **2 PCs and 1 Switch**, then connect them using **copper straight-through cable**



3. Assign IP addresses:
    - PC0 → `192.168.1.1`
    - PC1 → `192.168.1.2`



4. Observe packets in Simulation Mode and verify communication is successful



TCP (Transmission Control Protocol)

- Connection-oriented protocol (connection established before transmission)
- Reliable data transfer (guarantees delivery)
- Maintains correct order of data
- Uses acknowledgments (ACK)
- Retransmits lost packets

UDP (User Datagram Protocol) –

- Connectionless protocol (no connection setup)
- Unreliable data transfer (no guarantee of delivery)
- Does not maintain order of data
- No acknowledgments
- No retransmission of lost packets
- No flow control
- No congestion control